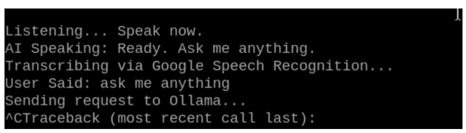
OLLAMA VOICE ASSISTANT DOCUMENTATION

The process to achieve a functional voice assistant was successful but highlighted some performance challenges. There were 3 core technical problems: first, fixing the Speech-to-Text (STT) component by installing flac and adjusting time limits to prevent the microphone from transcribing the AI's own output; second, resolving long Ollama (LLM) processing latency by setting a 120-second timeout; and third, overcoming repeated Git conflicts during the submission phase.

Another issue was the poor quality and challenging default male voice of the espeak Text-to-Speech engine. The default voice was often distorted and difficult to understand which required updating the TTS parameters in the Python script to use a female voice (en+f3) and adjust the rate and pitch to improve clarity. The female voice was better but still difficult to understand (see recording here)

Baseline AI Response:
The very first successful response revealed the model's default behavior. After the system accidentally transcribed its own "Ask me anything" prompt, the Phi-3:mini model generated a lengthy, boilerplate text detailing its privacy policy and ethical limitations. However, the response was too long and formal for a practical voice assistant.



User Response to Performance:
Functionally, the system later achieved a successful full-loop dialogue when the user asked, "What's the largest continent?". This confirmed that the local Phi-3:mini model could process and respond to a complex query. However, the critical takeaway from the testing phase was the user's response to latency. The long wait time (30+ seconds for the first query) while the model processed the request caused a little anxiety and a loss of confidence in the system. Future designs should prioritize providing immediate verbal feedback ("I'm thinking now...") to manage user expectations and mask the unavoidable processing delay for conversational outputs.